Status	Finished
Started	Thursday, 1 October 2026, 4:12 PM
Completed	Thursday, 1 October 2026, 4:15 PM
Duration	2 mins 51 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Searching for data in a list
b. Deleting data from a list
c. Inserting data into a list
d. Arranging data in a particular format

The correct answer is: Arranging data in a particular format

Question **2**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Bubble sort
b. Heap sort
c. Quick sort
d. Insertion sort

The correct answer is:
Insertion sort

Question **3**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 200,470,80,150,40,90,300,400,70,120
- b. 40,70,80,90,120,150,200,300,400,470
- c. 40,80,90,150,200,300,400,470,70,120
- d. 80,150,200,470,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **4**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. All of the above
- b. Searching
- c. Duplicates
- d. Selection

The correct answer is: All of the above

Question **5**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

- a. Quick Sort
- b. Bubble Sort
- c. Insertion Sort
- d. Merge Sort

The correct answer is: Bubble Sort

Question **6**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It makes the analysis slower
b. It has no effect on the analysis process
c. It complicates the analysis process
d. It allows for quicker identification of duplicates

The correct answer is: It allows for quicker identification of duplicates

Question **7**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are discarded
b. They are split again into smaller sublists
c. They are compared element by element
d. They are combined to form a single sorted list

The correct answer is: They are combined to form a single sorted list

Question **8**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Binary Search
b. Merge Sort
c. Bubble Sort
d. Linear Search

The correct answer is: Merge Sort

Question **9**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Greedy method
b. Backtracking
c. Divide and conquer
d. Dynamic programming

The correct answer is:
Divide and conquer

Question **10**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It requires external libraries
b. It is less efficient than custom sorting algorithms
c. It sorts data out of the box efficiently
d. It only works with integer arrays

The correct answer is: It sorts data out of the box efficiently

Question **11**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting makes it harder to search for items
b. Sorting decreases the efficiency of selection operations
c. Sorting is rarely used in programming
d. Sorting helps in finding duplicates quickly

The correct answer is: Sorting helps in finding duplicates quickly

Question **12**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

- a. It complicates the selection of items
- b. It slows down the process
- c. It makes the data unsorted
- d. It makes it easier to select items based on their relationship to the rest of the items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **13**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Linear Search
- b. Merge Sort
- c. Binary Search
- d. Bubble Sort

The correct answer is: Merge Sort

Question **14**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

- a. Using the divide-and-conquer approach
- b. Bubbling up the largest element to its correct position with each pass
- c. Being the most efficient sorting algorithm
- d. Sorting data in a non-sequential manner

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **15**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

- a. Each part is sorted independently
- b. The parts are merged without sorting
- c. Each part is searched for a specific element
- d. The parts are ignored

The correct answer is: Each part is sorted independently